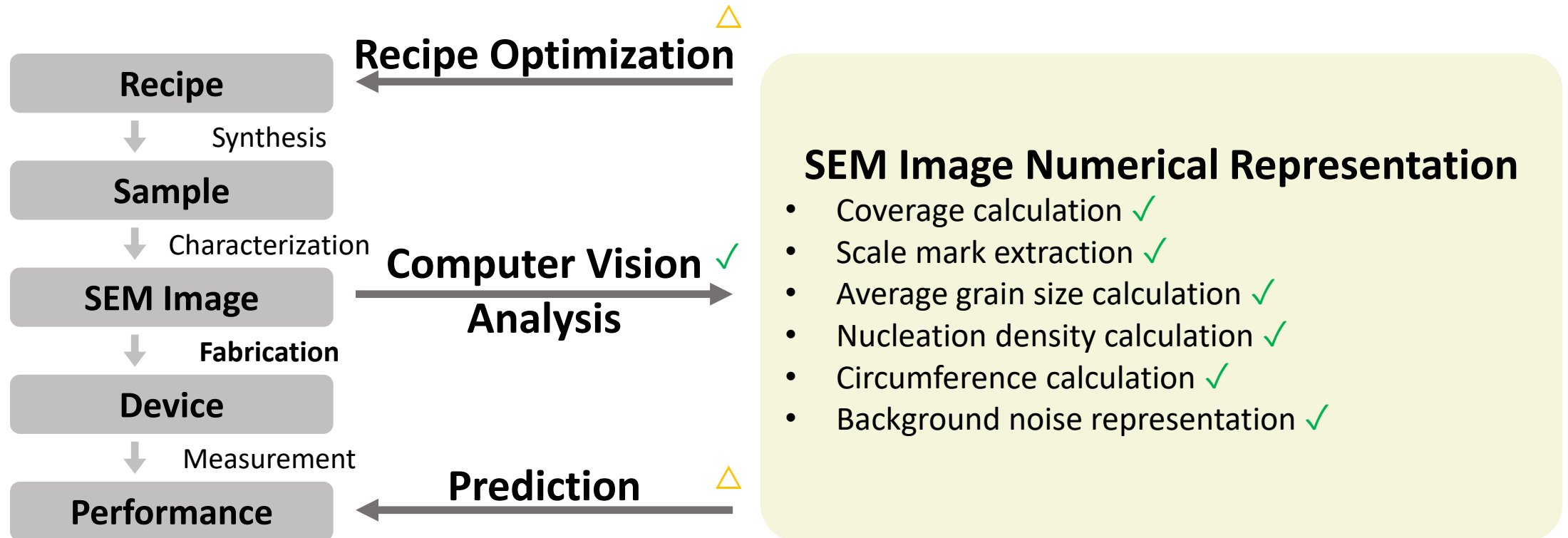


Machine-Learning-Driven Numerical Analysis of SEM Images for Optimizing MOCVD-Grown MoS₂ Device Performance

Collaboration - 02

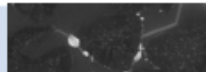
Meng-Chi (Ed) Chen

NMEG, 2025-11-23



1 hr

3 hr

Batch	Percentage of water	Lateral growth rate	Shape	Batch	Coalesce condition	Add layers	Mobility	
619	100%	6,000 nm/hr		621	Poor		Much	Good
634	50%	2,000 nm/hr		623/625	Good		Much	Good
627	30%	1,000 nm/hr		633	Poor spin coating			
626	10%	600 nm/hr		632	Good		Few	Good
624	3% (previous)	500 nm/hr	Δ 	628	Good		Even less	Poor
629	1%	50 nm/hr	Δ 	/	/	/	/	/
1"	Evaporation	1,000 nm/hr	Δ		Good	Even less	Good	
8"	Evaporation	500 nm/hr	Δ		Good	No	Good	
8"	Extreme	20,000 nm/hr ?	Δ		?	?	?	
8"	Low-T (3% eq)	200 nm/hr	Δ		Good	No	Good/Poor	

Ignore

NA.

How to describe an SEM image of a 2D material?

- **Coverage:** Percentage of area covered by crystal. (Unit: %)
- **Grain size:** Size of individual crystal. (Unit: μm^2)
- **Nucleation density:** Number of crystal in unit area. (Unit: $\frac{ct}{\mu m^2}$)
- **Shape:** Shape of crystal.
- **Impurity:** Precursor residuals or other small crystals.
- **Orientation:** Direction of crystals.
- ... More

How to get these information?

- ImageJ (or other scientific software for microscopic image processing)
 - Require multiple steps of settings and adjustments.
 - Some of our desired function are not exist (or require excessive post process).
- Python
 - Customize the field-specific needs.
 - Streamline the process.

Execution Steps

1. Extract scale bar info from SEM image.
2. Convent SEM image to grayscale.
3. Crop out the center square.
4. Enlarge image for better CV processing (cv2.INTER_LINEAR).
5. Slightly blur the image (cv2.medianBlur(k=5)).
6. Enhance image (CLAHE and merge).
7. Binarize image with Otsu's method.
8. Calculate coverage and create histogram.
9. Watershed image.
10. Segment image and save segmentations.
11. Calculate segmentation result.
12. Save result as excel.

Numerical Representation

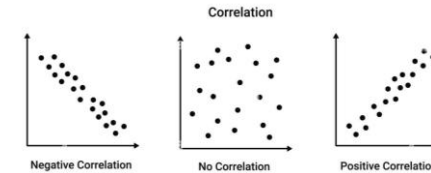
- Pixel per um (sclbar_text, pixel_length_sclbar, pixels_per_um).
- Coverage (pct_black, pct_white).
- Segmentation (cnt_seg_success)
- Nucleation density (nd_per_sqnm)
- Grain size (avg_gs_squm, std_gs_squm)
- Circumference (avg_circumf_um, std_circumf_um)
- Cricumference square to area ratio (avg_csar, std_csar)
- Background noise (avg_bkgd_pixel, std_bkgd_pixel)

How are these measurements relating to device performance?

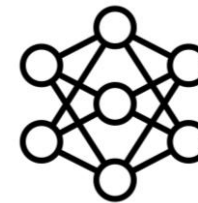
Numerical representation of 2D material SEM images:

- Pixel per um (sclbar_text, pixel_length_sclbar, pixels_per_um).
- Coverage (pct_black, pct_white).
- Segmentation (cnt_seg_success)
- Nucleation density (nd_per_sqnm)
- Grain size (avg_gs_squm, std_gs_squm)
- Circumference (avg_circumf_um, std_circumf_um)
- Circumference square to area ratio (avg_csar, std_csar)
- Background noise (avg_bkgd_pixel, std_bkgd_pixel)

It might be harder to see the relation of one single measurement relates directly to the device performance.



ML model can help correlate measurements from SEM image to device performance when there is more data.



Device Performance

Demo Video

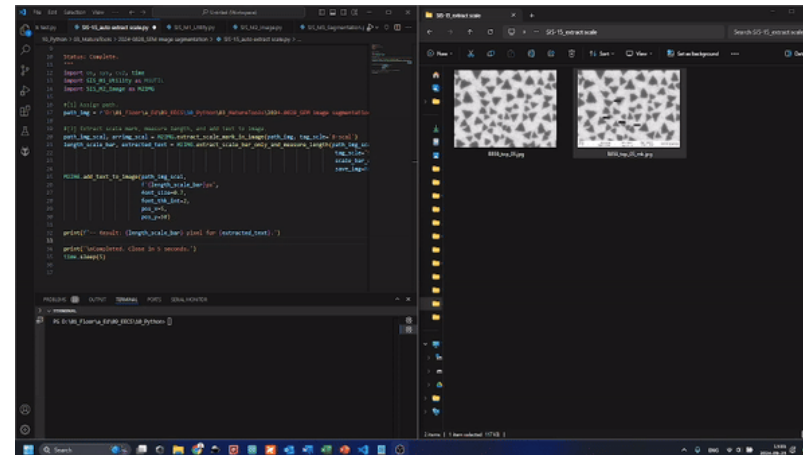
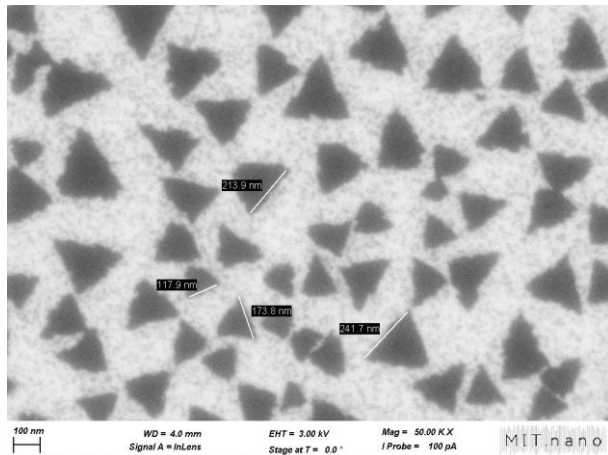
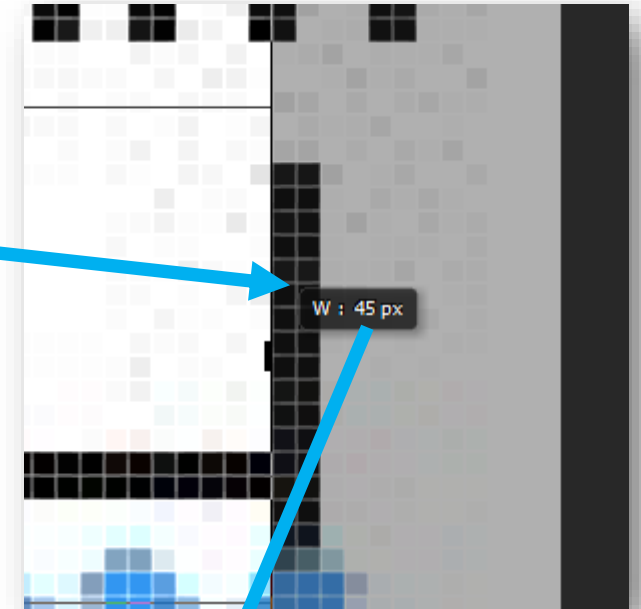
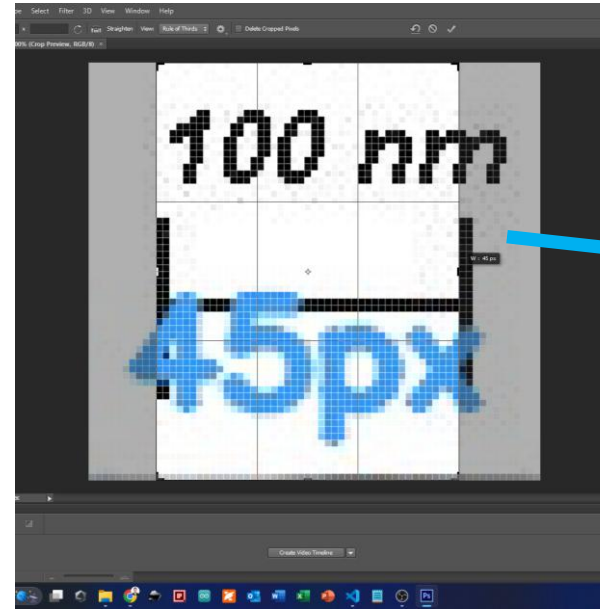
- Single folder SEM image segmentation.
- Took 2 mins to process 4 images.

[\[OneDrive\] Image Seg Demo 480p.mp4](#)

[\[Local\] Image Seg Demo 1080p.mp4](#)

Auto Scale Bar Extraction

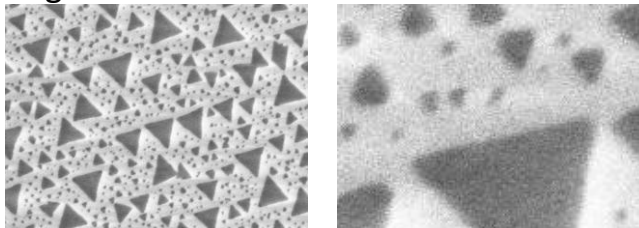
- Average pixel in columns and rows to cut out desired area of scale bar.
- Use `pytesseract` to read the number of the scale bar.
- The length of scale bar matches the actual pixel checked with Photoshop.



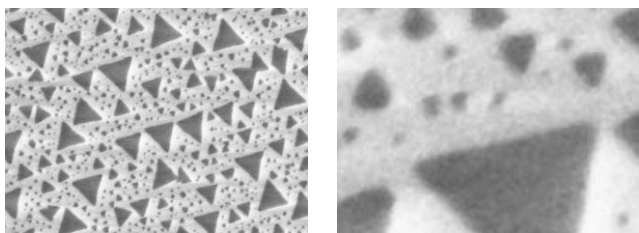
```
[extract_scale_bar_only_and_measure_length] Processing: B859_top_05_mk (8-scal).jpg
[arr_col_mean]
253 253 253 253 254 254 253 251 251 142
Scale bar boundary
142 236 230 223 203 212 232 244 244 219 208 223 228 224 208 220 244 220 208
223 227 223 207 219 243 243 242 243 243 231 211 223 235 235 223 211 227 243
230 210 221 234 234 222 110 125
Scale bar boundary
242 243 231 219 235 251 251 253 253 253
-- Scale bar = 45 pixel for 100 nm.
-- Result: 45 pixel for 100 nm.
Completed. Close in 5 seconds.
PS D:\01_Floor\B859\00_EECS\10_Python>
```

Coverage

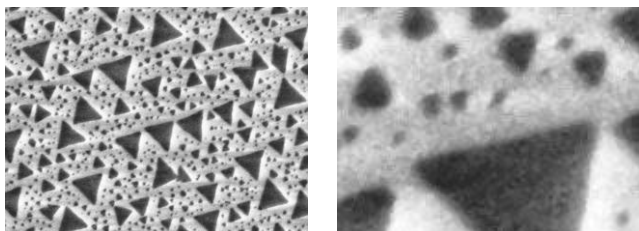
Original



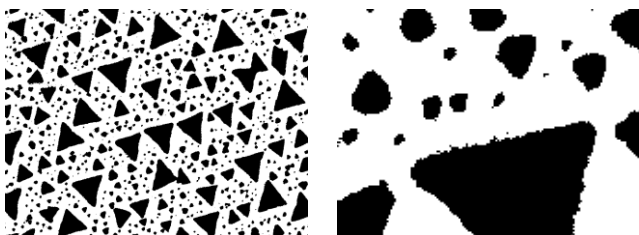
Gentle Blurred



Gentle Enhanced



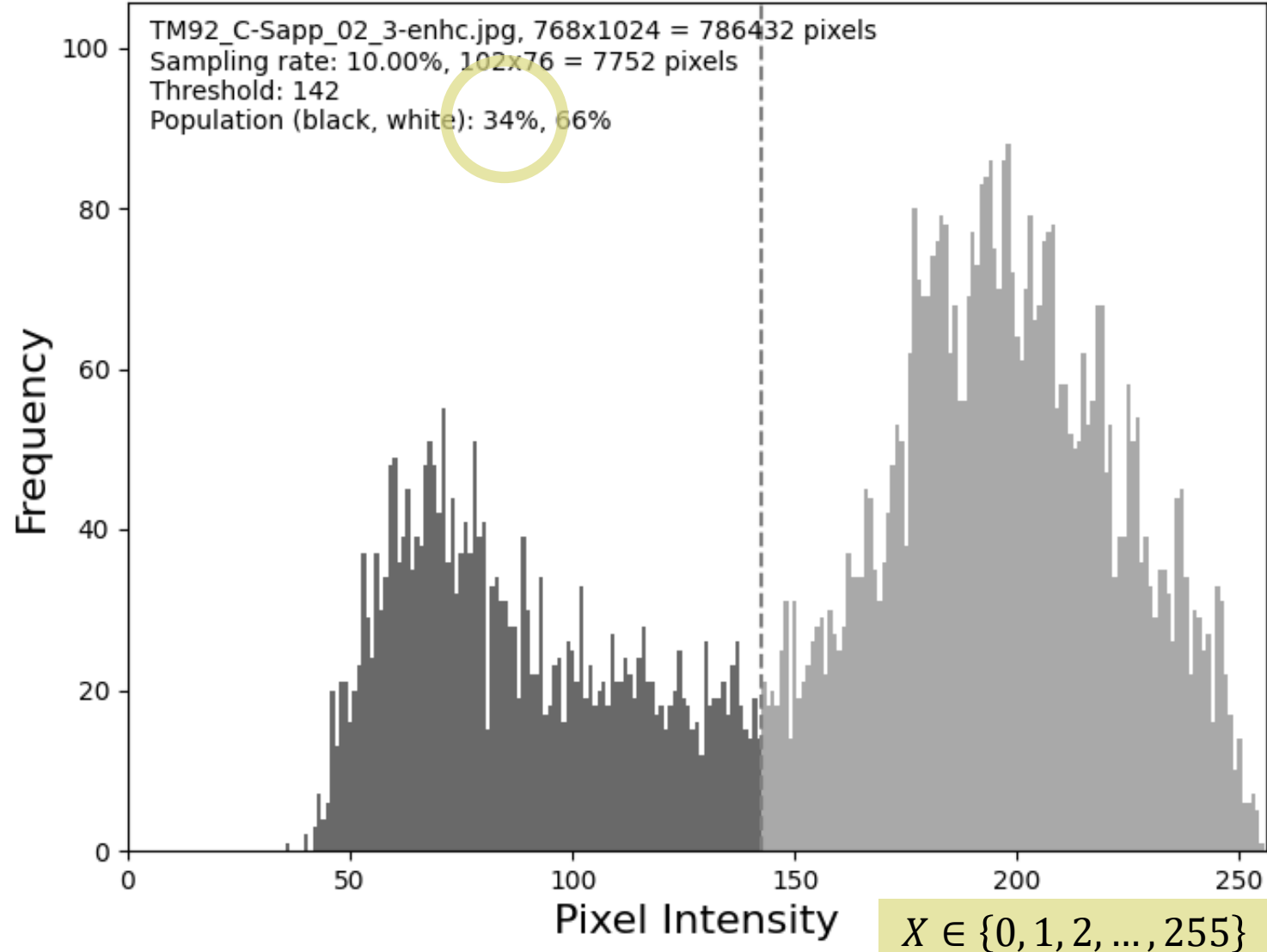
Binarized



Ed: TM92_C-Sapp_02

2025-11-23

Intensity Histogram



Otsu's method of thresholding: minimizing the intra-class variance.

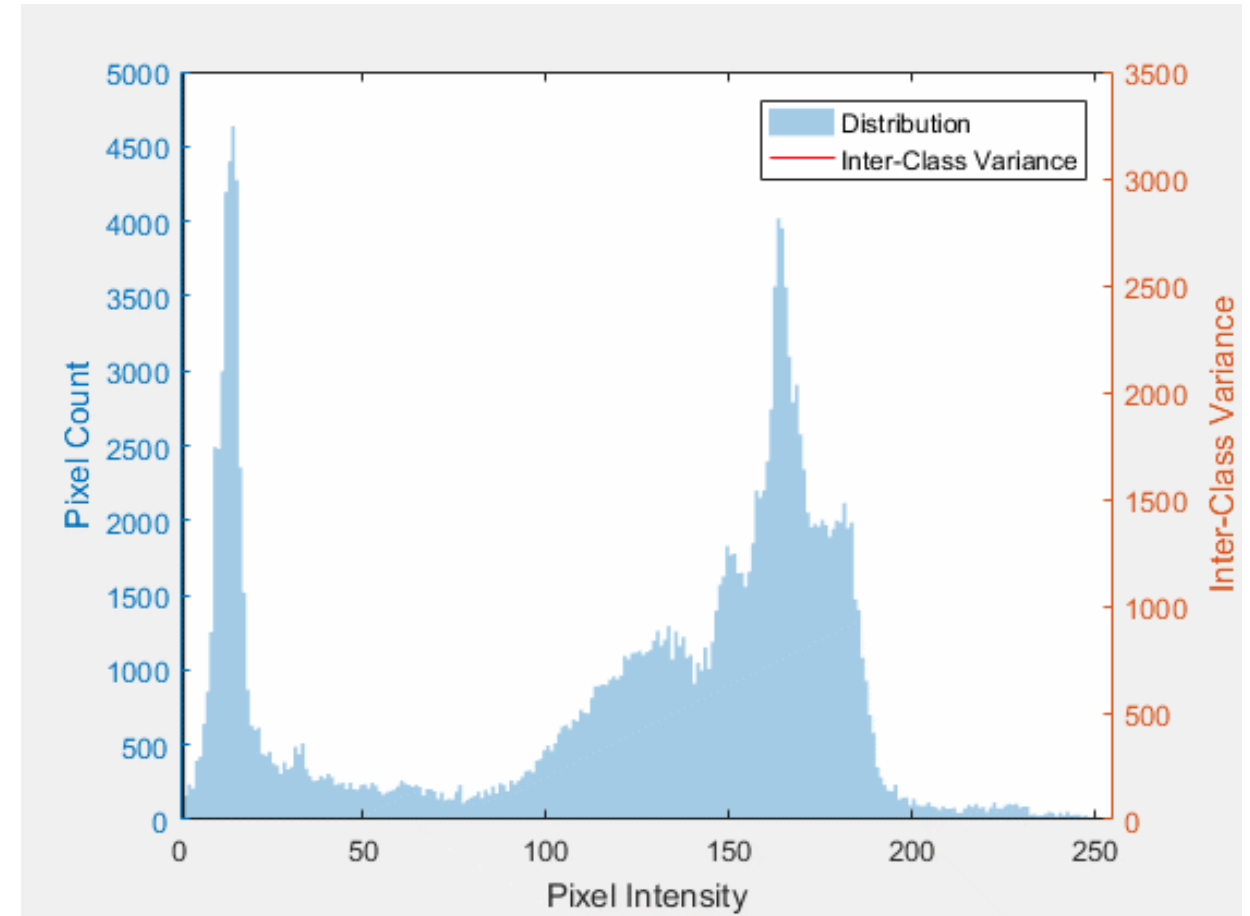
edchen93@mit.edu

10

Otsu's Method

$$T_{opt} = \arg \max_T \sigma_B^2(T)$$
$$\sigma_B^2(T) = \omega_0(T)\sigma_0^2(T) + \omega_1(T)\sigma_1^2(T)$$

- Otsu's method selects a threshold T_{opt} that **maximizes the between-class variance** $\sigma_B^2(T)$. (Or equivalently, minimizing intra-class intensity variance.
- $\sigma_B(T)$ and $\sigma_B^2(T)$ are standard deviation and variance between groups, respectively.
- $\omega_0(T)$ and $\omega_1(T)$ are proportion of pixels in background and foreground respectively, and $\omega_0(T) + \omega_1(T) = 1$.



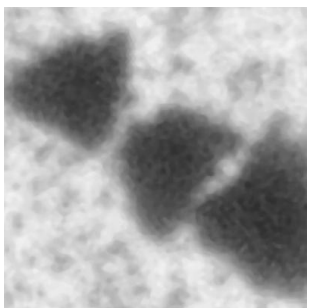
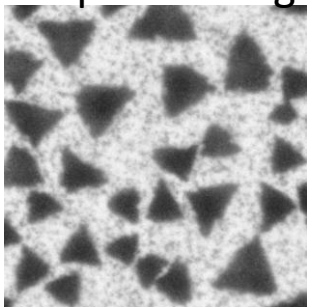
```
otsu_threshold, arrimg_5bnrz \  
    = cv2.threshold(arr_img, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)
```

Watershed

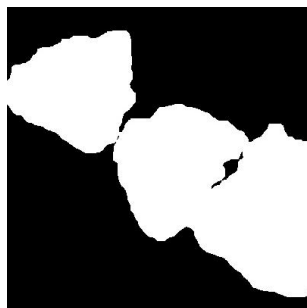
Cut through crystals that sticks together.

1. Prepare image: grayscale, noise reduction, then binarize.
2. Identify **background** by dilating (extend edge a little bit) the binary image.
3. Apply **distance transform** that turn none-zero pixel into the distance to nearest black pixel.
4. Set threshold to make out **foreground** from the distance transform result.
5. Subtract foreground from background to create **unknown regions**
6. Apply **watershed** by flooding between foreground islands to cut though unknown regions.

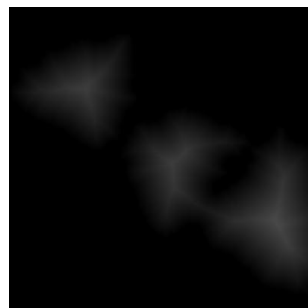
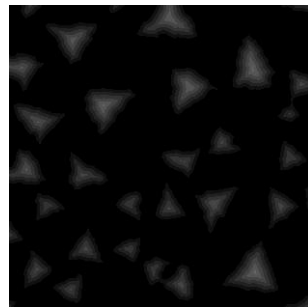
Prepared image



Background



Distance Transform



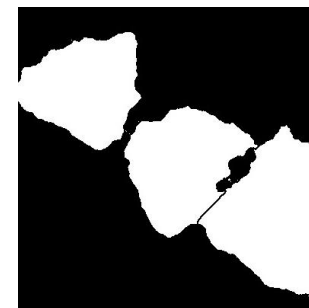
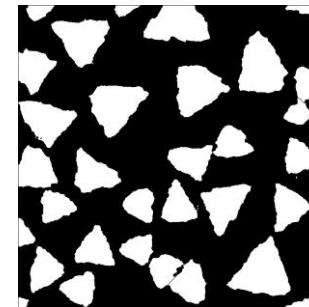
Foreground



Unknown



Watershed



Distance Transform

Distance Transform applied to a binary image so that:

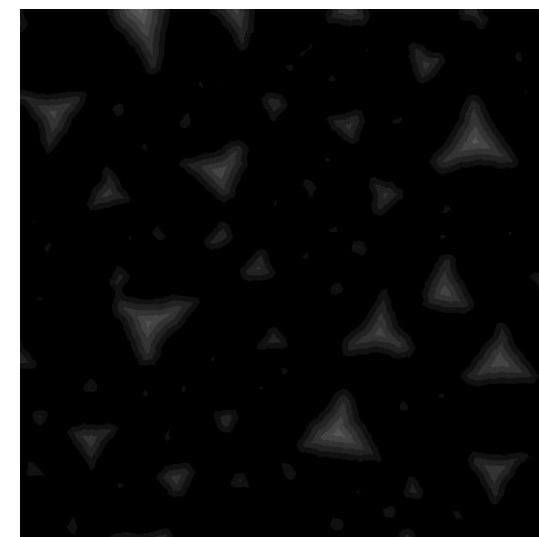
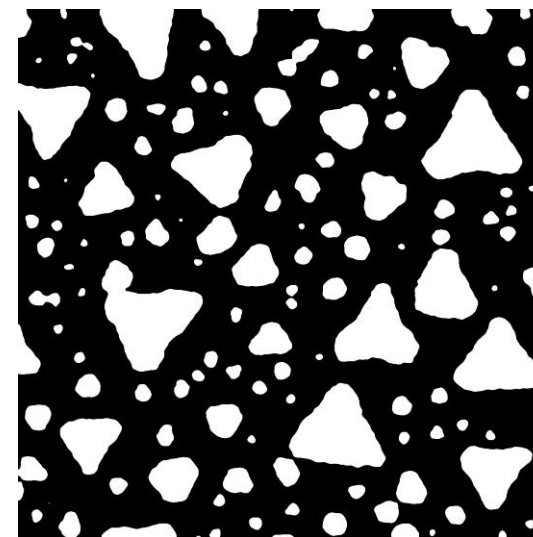
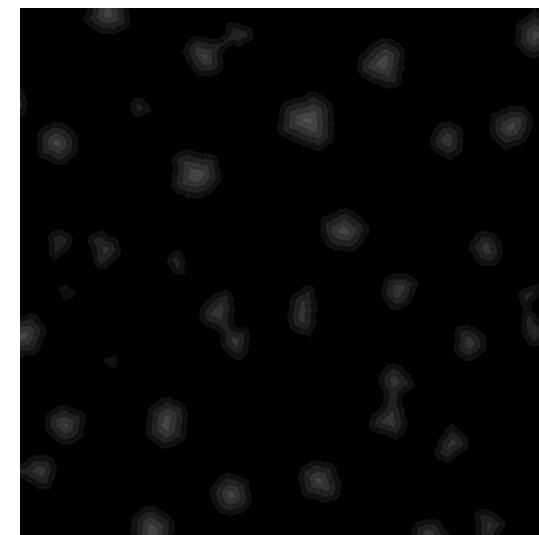
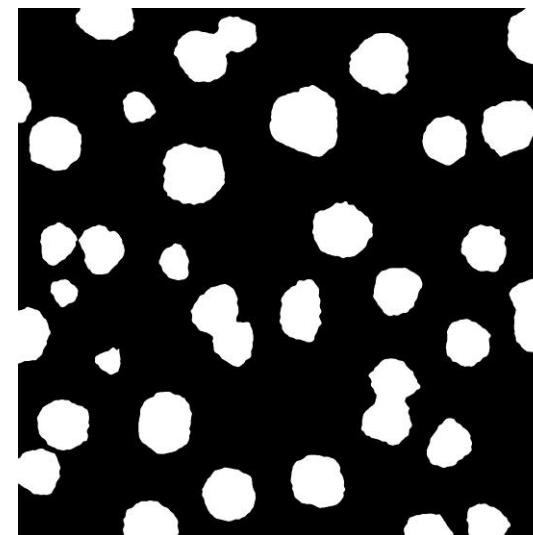
1. Black (0) regions remain black, representing the background.
2. White (non-zero) regions are transformed so each pixel's value represents the distance (in pixels) to the nearest black (0) pixel.

0	0	0	0	0	0	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	0	0	0	0	0	0

Binary Image

0	0	0	0	0	0	0
0	1	1	1	1	1	0
0	1	2	2	2	1	0
0	1	2	3	2	1	0
0	1	2	2	2	1	0
0	1	1	1	1	1	0
0	0	0	0	0	0	0

Distance transformation

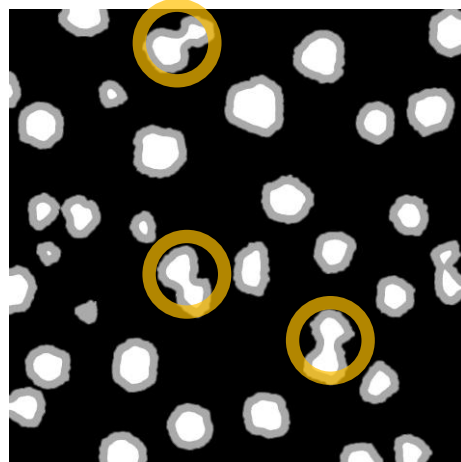


Input image

After distance transform

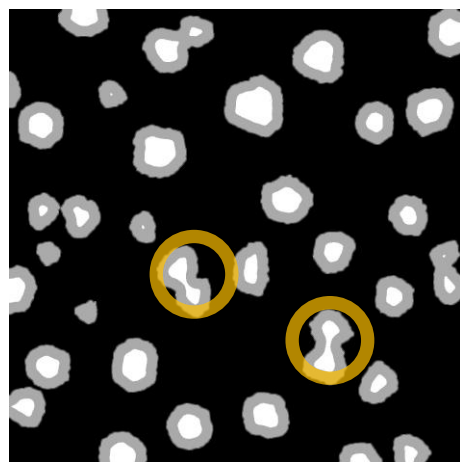
Watershed Parameter Adjustments

- `frgd_thrs_ratio` is a parameter between 0 and 1 that controls the erosion from the edge of foreground.

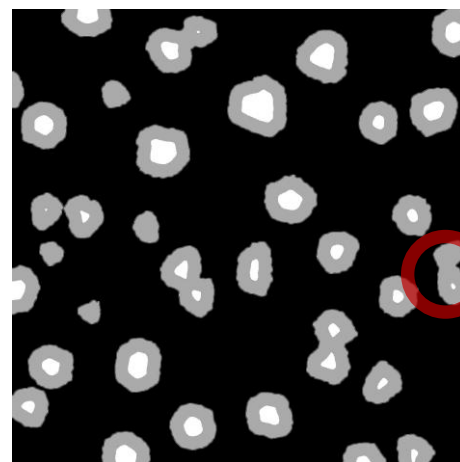


0.3

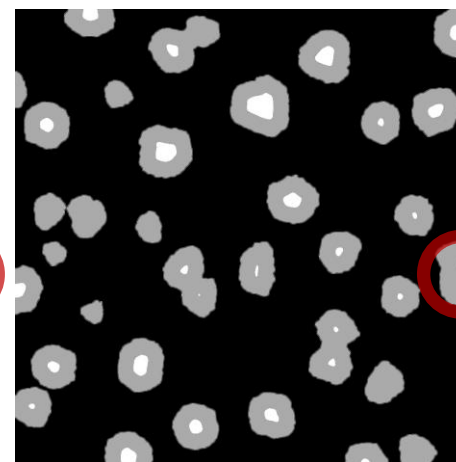
Too small, some foregrounds are linked together.



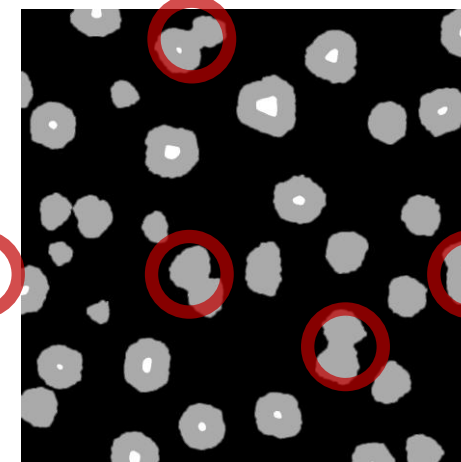
0.4



0.5

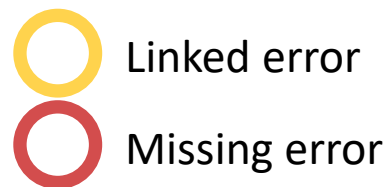


0.6



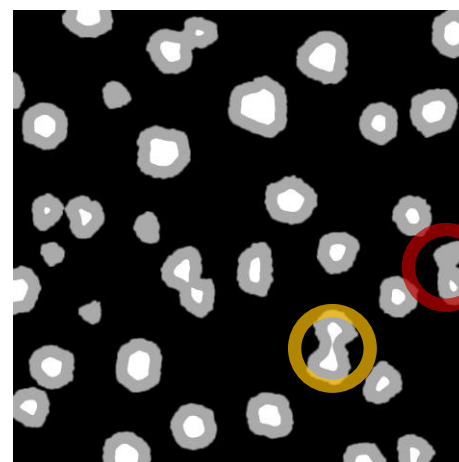
0.7

Too high, some foregrounds are missing.



Linked error

Missing error

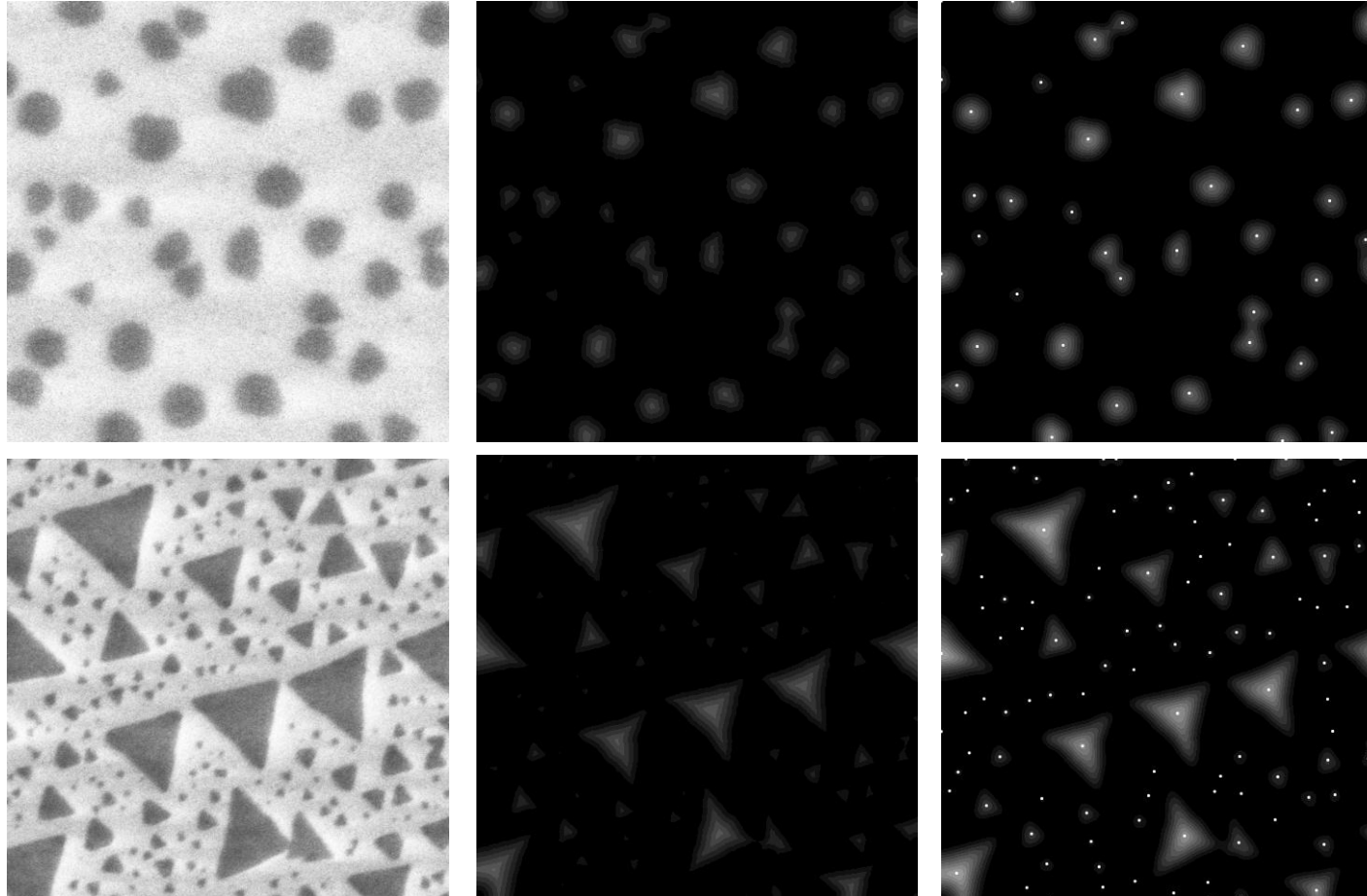


0.45

- It's hard to find a middle ground that both error won't occur.
- Setting suitable for this image might not be suitable for another image.

Watershed Improvement — Add foreground at local maxima

- Move a searching window through the image to find local maximum.
- If a maximum is found, mark it with white, so that it will be recognized as a foreground (island) in the later watershed.



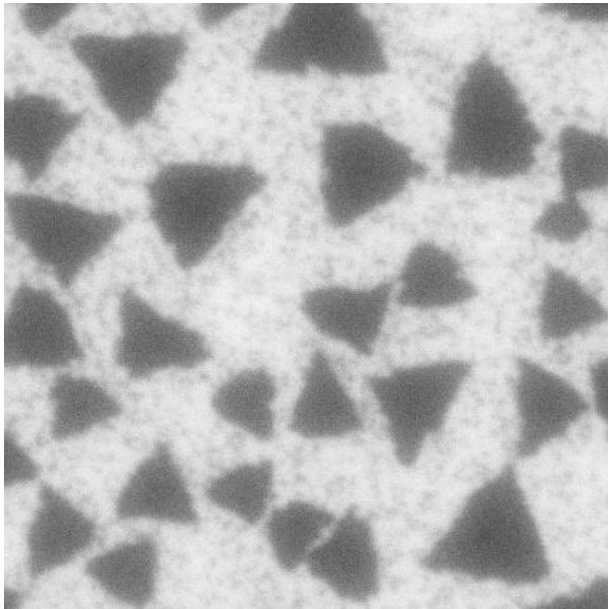
Original

After distance transform
edchen93@mit.edu

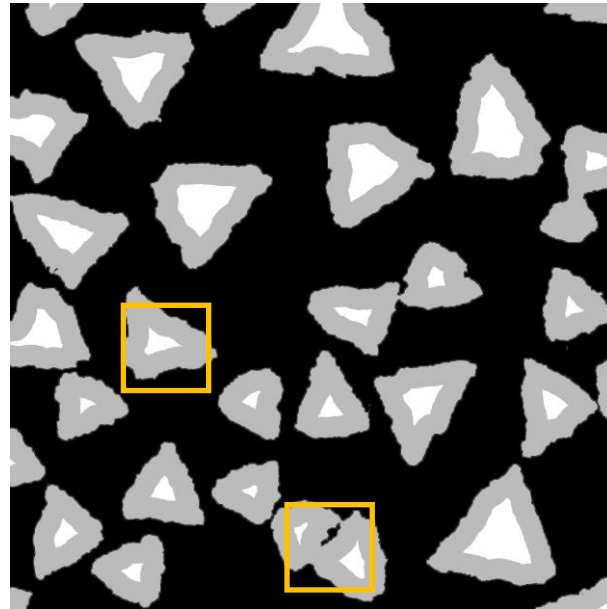
Mark local maxima

Foreground Sharp Edge Smoothing

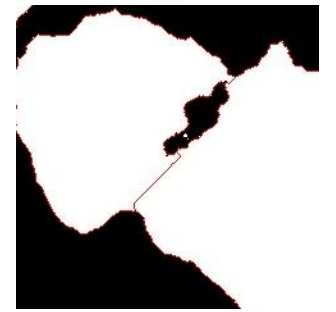
- Some crystal have sharp edge that during foreground extraction, a single crystal will become multiple crystals, causing error in the later watershed.
- Solution: Applied edge smooth to the extracted foreground, then proceed watershed.



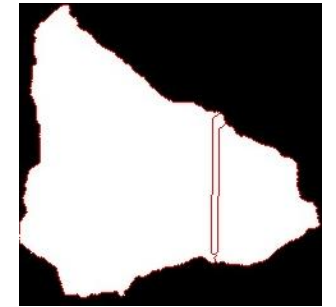
Input



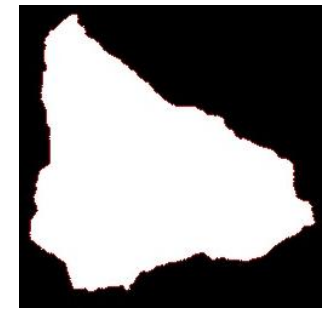
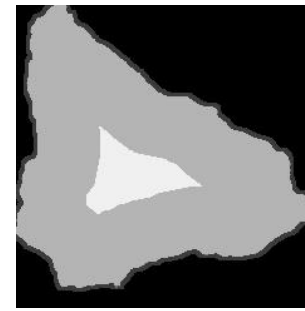
Foreground/Background overlay



Normal case.



This issue case.

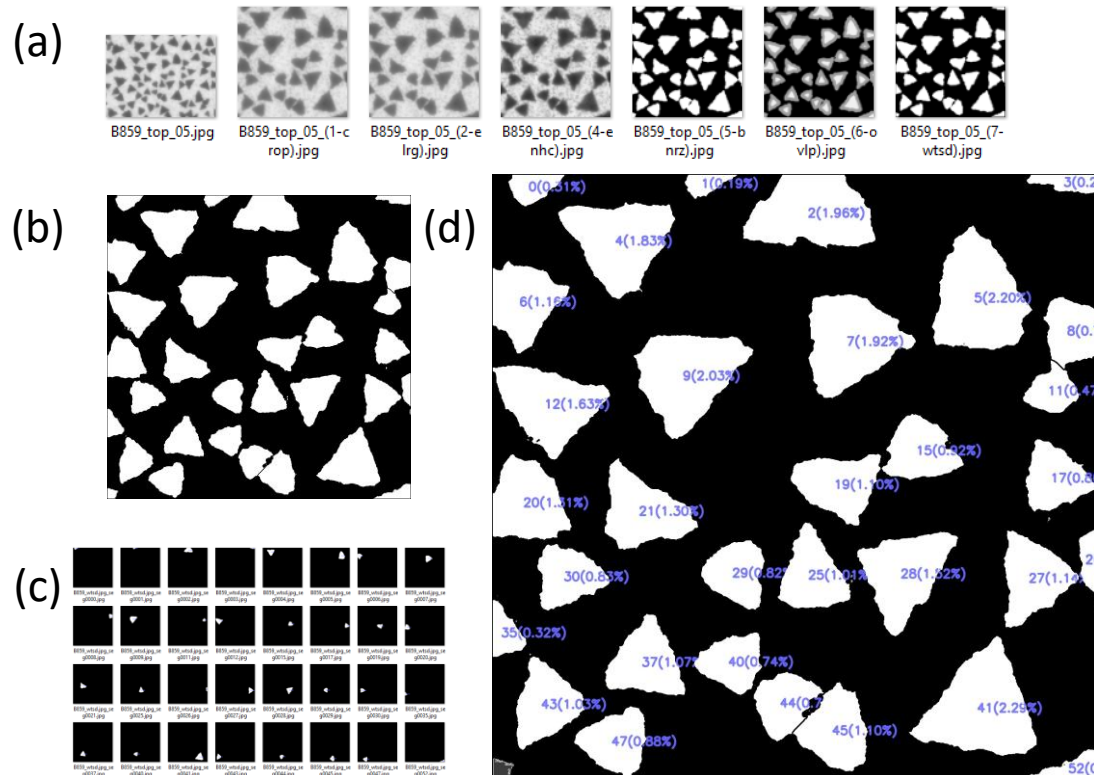


Corrected result.

Segmentation

Approach with more reliable traditional segmentation process.

- (a) Image process: Crop, enlarge, blur, enhance, and watershed.
- (b) Before segmenting (last of image process).
- (c) Segmentation result (layer by layer).
- (d) Segmentation result (overlap).
- (e) Precessing terminal



(e)

```
Removing unreasonably small segmentation. [S051] 11 pixel (0.000%).
-- [S052] 7001/2250000 = (0.31%)
-- [S053] 2079/2250000 = (0.09%)

Removing unreasonably small segmentation. [S053] 2079 pixel (0.092%).
-- [S054] 3/2250000 = (0.00%)

Removing unreasonably small segmentation. [S054] 3 pixel (0.000%).
-- [Nucleation density] 1.422e-05 (crystal / pixel^2)
-- [Average Grain size] 72387.469 (pixel^2)
-- Overlapped image saved to D:\01_Floor\A_Ed\09_EECS\10_Python\03_MatureTools\2024-0828_

-- [Nucleation density] 25.920 (crystal / um^2)
-- [Average Grain size] 0.040 (um^2)

save_df_as_excel_overwrite]
--Dataframe saved to D:\01_Floor\A_Ed\09_EECS\10_Python\03_MatureTools\2024-0828_

[adjust_multiple_column_widths]
--Column 'C' in sheet 'seg' has been adjusted to width 20.
```

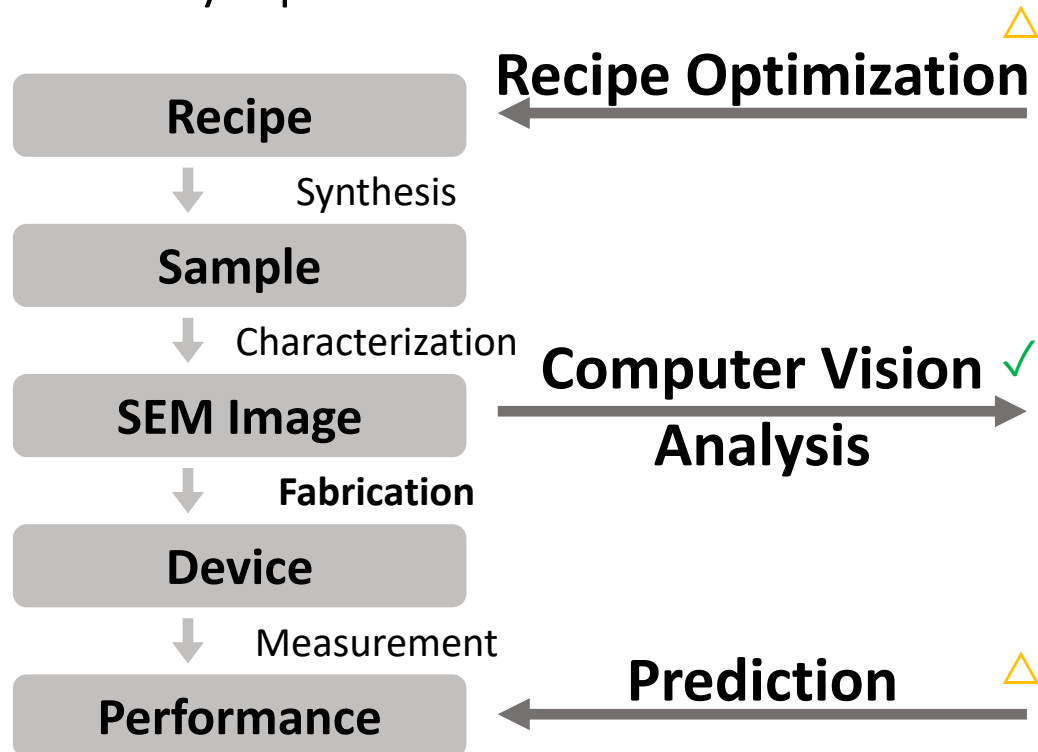


(g)

	list	success	pos_x	pos_y	area	at_edge
0	0	0	100	31	20796	TRUE
1	1	1	527	23	13158	TRUE
2	2	2	791	96	132549	TRUE
3	3	3	1424	20	17871	TRUE
4	4	4	314	165	123567	FALSE
5	5	5	1203	306	148821	FALSE
6	6	6	77	316	78501	TRUE

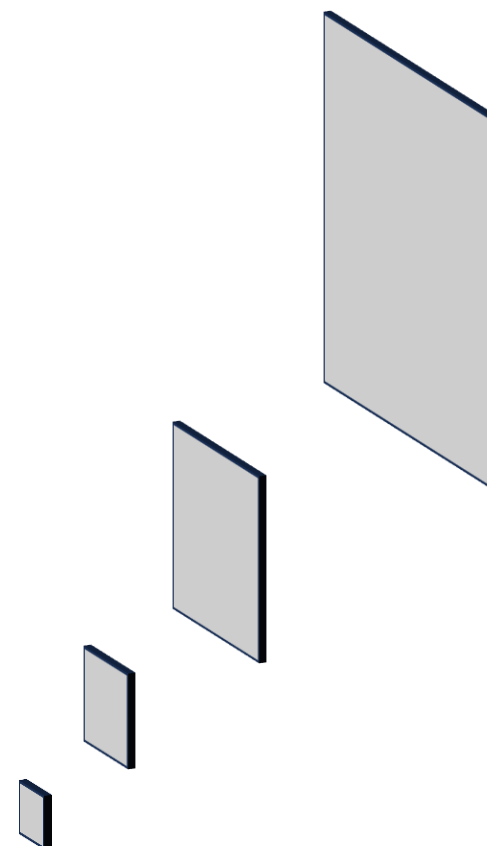
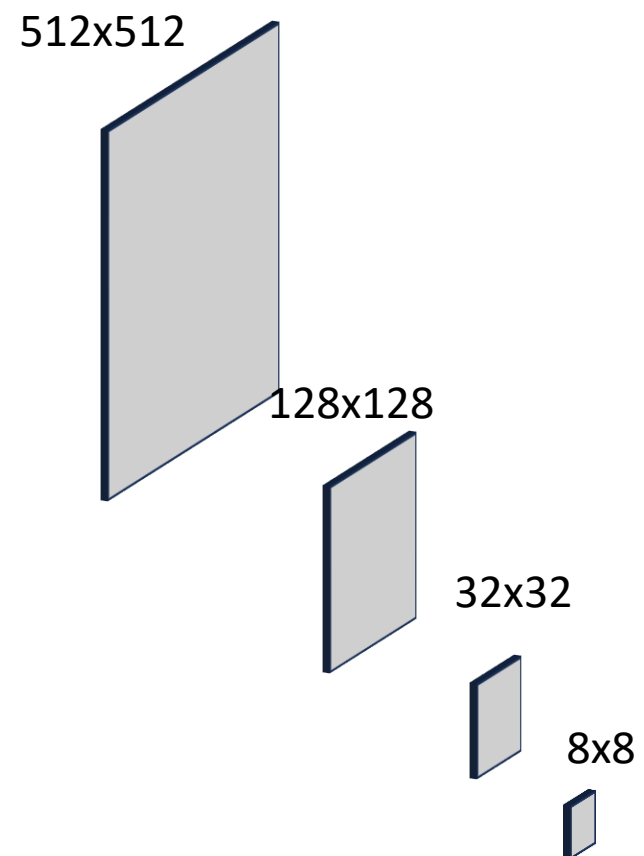
Future Work

- Improve the watershed function to better adapt to SEM images in various condition.
- Obtain device data from Jiadi or Yixuan to correlate the numerical representations of SEM images with device performance.
- Conduct literature research on the feasibility of using machine learning models to embed SEM images into dense array representations.



SEM Image Numerical Representation

- Coverage calculation ✓
- Scale mark extraction ✓
- Average grain size calculation ✓
- Nucleation density calculation ✓
- Circumference calculation ✓
- Background noise representation ✓



Future Work

Calibri Light, 20-24 Normal

- Crystal Violet
- Patent
- Mobility data (B620 ~B640)

Future Work

Calibri Light, 20-24 Normal

- Crystal Violet
- Patent
- Mobility data (B620 ~B640)